BOOKLET NUMBER: 48333

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VARDHAMAN
COLLEGE OF ENGINEERING
Shamshabad – 501 218, Hyderabad

CAT EXAMINATION ANSWER BOOKLET

VARDHAMAN COLLEGE OF ENGINEERING
23881A05EC(G)
A8604 23.10.2025 FN

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CAT - I / II EXAMINATION I / II / III / IV B.Tech / M.Tech / MBA I / II Semester 10/2025
(Strike off whichever is not applicable) (Month and Year)

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Branch : Computer Science and Engineering

Subject : Web Technologies

Subject Code : A8604

Date of Exam : 23-10-2025

23/10/25
Signature of the Invigilator with date

(Booklet contains 20 pages)

INSTRUCTIONS TO THE CANDIDATES

1. This booklet contains 20 pages. Candidates must ensure it before writing and in case a defective answer book is issued it must be returned to the invigilator and a new and defect free booklet must be obtained.
2. Before the candidate begins to answer, registered number, particulars of year, semester, subject, etc., are to be filled in. Failure to enter all or any of these particulars may disqualify the paper from valuation.
3. Candidate is prohibited from
   (a) writing:
   - anything addressing the examiner in any manner whatsoever, in their answer book.
   - objectionable/obscene language in the answer book.
   - anything other than their Registered No. on the question paper.

   (b) either seeking or providing any assistance to the fellow candidates in the exam.

   (c) possessing a manuscript or a printed matter, in any form, in the examination hall.

   (d) bringing loose sheets or paper into the examination hall and detaching any paper from the answer book.

   (e) carrying Mobile Phone, Smart Watches and any other Electronic Gadgets etc., to Exam Hall.

Violation of these instructions will be viewed as a case of malpractice, which is a punishable offence.

4. Before beginning to answer any question, candidates must write the correct question number, and should not write
5. Answers must be written legibly on both sides of the paper. There shall be about 25 lines in each page. It is not necessary to begin each answer on a fresh page. Candidates should not use any other ink, except BLACK or BLUE ink.
6. Rough work, if any, must be separated, from the subject matter, by a line and noted as rough work.
7. The answer book, at the end of the examination, must be handed over to the Assistant Superintendent (Invigilator) by the candidate.

This responsibility lies with the candidate only.

8. Candidates should maintain absolute silence during the time of examination. Misbehavior, in any form, by the candidate, in the examination hall, will attract severe punishment.
9. Candidates are permitted to leave the examination hall only after the allotted time.
10. **No additional answer books will be supplied.**

To be filled in by the Examiner only

MARKS AWARDED

PART – A				PART – B						
Q No.1				Q No.	2	3	4	5	6	7
a		f		a						
b		g		b						
c		h		c						
d		I		d						
e		j		Total						
Total (a to j)										

In Words:

In Figures:

Signature of the Examiner

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START WRITING FROM HERE

Q.No.

PART-A

1) a) → Class.forName() is used to load the database driver.
It loads the databases driver into system so that we can establish a connection.

Syntax:

Class.forName("com.mysql.cj.jdbc.Driver");

1) b) → Different interfaces for SQL queries are:

• ExecuteUpdate(): used to perform/execute DDL commands, like insert, alter, drop, etc...

• executeQuery(): used to execute select statements mostly.

Syntax:

stmt.execute~~Update~~Query("select * from students");

stmt.executeUpdate("Create table coffee(cof_name varchar2(30), cost INTEGER)");

1) c) → ResultSet interface is used to display result from a database.
→ It is used mostly to select statement Queries.

Syntax:

~~// ResultSet rs = stmt.getResults(~~

ResultSet rs = stmt.executeQuery("Select * from Students");

~~while (rs.next()){~~
~~// Integer String id=~~
~~int id= Integer.parse~~

→ It stores data in "rs" which will be displayed manually using commands

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Q.No.

1) d) → interfaces that handle HTTP requests are:

HttpServletRequest : Handles request side

HttpServlet Response: Handles response side

HttpSession: Handles session applicns

Ex:

```
public void doGet(HttpServletRequest req, HttpServletResponse res)
        throws IOException, servletException {


}
```

1) e) → destroy() method in servlets is called when the need of servlet is finished and it needs to be closed.

→ when destroy() method is called, it stops all the connection, and closes the servlet.

1) f) → Cookie is sent from servlet to browser requesting some data. This is again sent back from browser in next request.

→ ~~Ef~~ → It is sent using method response.addCookie("");

Syntax/Ex:

```
Cookie user = new Cookie ("username", "sathvik");
user.setMaxAge (60*60*24); //1-day
response.addCookie("user");
```

Q.No.

1) g) → JSP tag, used to declare Java method, one way is using directive 'page' and including language 'java'

<%@page language = java %>

→ another is denoting the java code inside scripting tag.

<% %>

→ To declare java methods and objects, we use.

<%! %>

Ex: <%! int var = 10; %>

1) h) → include directive in JSP tells the web container to include required file (or) package.

<%@ ~~page~~ include file = "main.txt" %>

→ include action element is an inbuilt function to define some predefined function.

<jsp:include file = "java1.java" />

1) i) → to delete a cookie, we use setMaxAge() method. we set the maxage of cookie to '0' (zero) which deletes the cookie.

Syntax:

```
cookie var = new cookie ("user", "john");
var.setMaxAge (0*0*0);
response.addCookie (var); //deletes cookie
```

Q.No.

1) j) → Diff types of drivers available are:

i) JDBC - ODBC / Type-1 / Universal driver

ii) Native API driver / Type-2 driver

iii) Network Protocol / Middleware driver / Type-3

iv) Thin driver. / Type-4 driver.

→ for datatabase specific drivers, we use

- Network Protocol / Type-3 driver which contains all drivers in it. used when to perform operns on multiple databases
- Thin driver is used when we need to perform on a single database.
- If type-3 & 4 not available, use type-2
- else, use type-1 (if none of above types available).

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PART-B

2) Steps involved in establishing database connection are:

i) Load Driver manager
ii) Connection Establishment
iii) Statements prep & Execution
iv) Viewing results
v) disconnecting connection.

i) Load Driver manager:

- Loading driver is important to establish connection with Database.
- Driver is loaded using:

```
Class.forName("com.mysql.cj.jdbc.Driver");
```

- After loading driver using Class.forName() method, we establish connection.

ii) Establishing connection:

- Connection is established by DriverManager class and getConnection() method.

```
Connection con = DriverManager.getConnection("jdbc:mysql:
        //localhost:3306/studentdb","root","admin123");
```

- Now, connection is established. we prepare and execute stmts.

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iii) Executing Stmts:

→ statements are created using

Statement stmt = con.createStatement();

→ statements are executed using executeUpdate() method.

stmt.executeUpdate("Create table students (Studentid INTEGER, studentname varchar2(30))");

stmt.executeUpdate ("Insert into students values (101, "Anil")");

iv) Result checking:

→ Results are checked/display using ResultSet.

ResultSet rs = stmt.executeQuery("Select * from Students");

→ results are displayed using executeQuery(), used mostly for select statements.

→ The data stored in rs is ~~stored~~ displayed manually.

```
while (rs.next())
{
    int id = ~~rs.get~~ req.getParameter ("student id");
    String name = req.getParameter ("studentname");
    out.print( id + " | " + name);
}
```

v) Closing connection:

→ closing connection is an important step.

→ It is done using

con.close();

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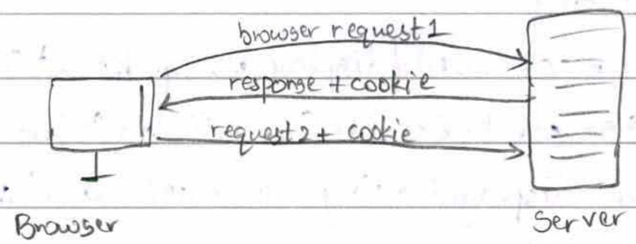
4) → Cookies are used to store small size of data (<= 4kB).

→ Cookies are sent by server to web browser to collect user data.

→ When a browser makes a request, server sends a cookie along with response to browser.

→ Browser then gives the required info to the cookie and sends it back in consequent requests.



→ Types of cookies:

i) non-persistent Cookie: This cookie is present for short-time, like until user closes the window, etc.

ii) persistent cookie: This cookies stay for long-time, until user signs-out, or logs-out.

→ Features:

→ Cookies are stored on client side browser.

→ Cookies cause load to client.

→ Cookies store very short amount of data.

→ Cookies are prone to security breach; data like passwords should not be stored in cookies.

addCookie
getCookies
~~setAttri~~
get Int
getString
setMaxAge

adv

dis
load to client
short stor
less secure

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→ Functions of cookie:

addCookie(): adds a cookie to the browser.
getCookies(): gets cookies from the browser.
getInt(): gets integer val of mentioned category.
setMaxAge(): sets maximum age of cookie
getString(): gets string value of mentioned category.

Ex: Below is an example to create a cookie, and retrieve it, and delete it.

```
import java.io.*;
import java.servlet.*;
import java.servlet.http.*;
~~pd~~ class cex extends HttpServlet {
        public class doGet(HttpServletRequest request, HttpServletResponse
response ~~resp~~) throws ServletException, IOException {
    cookie user = new cookie("username", "John");
      user.setMaxAge(60*60*24);
    response.addCookie(user);
    Cookie [] ck = request.getCookies();
        for (Coookie c: ck)
              String name = ~~user~~ ck.getString("username");
          out.println(name);
    user.setMaxAge(0);
```

Q.No.

Lifecycle of Servlet

5)

→ A servlet is a program which runs javacode on webservers like Tomcat.

→ Different steps in lifecycle of servlet are:

i) loading servlet

ii) Initialization (init())

iii) Service (service())

iv) Destroy (destroy())

i) when the user makes a domain request, the browser roots it to http and sends http request to the server. The server roots the http request to the matching/required servlet and sends the servlet to the browser. This servlet is now stored in browser address.

ii) After sharing the servlet to the browser, the server invokes the "init()" which starts the server. init() is called only once when the servelet is to be initiated.

Syntax:

```
public void init() extends ServletException {
    - - - -
}
```

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iv) whenever, the ~~server~~ browser makes a request, the server calls service() which then the servlet responds. The service() is called as many times as the browser makes request. Based on the request made, if it is a GET request, doGet is called. If it is a POST request, doPost is called. Many other methods are delete, retrieve, etc...

<u>Syntax</u>:

```
public void doGet(HttpServletRequest req, HttpServletResponse res)
    throws ServletException, IOException{
        - - - -
    }

public void doPost(HttpServletRequest req, HttpServletResponse res)
    throws ServletException, IOException{
        - - -
    }
```

v) After all the requests are done, or when the browser no longer needs the servlet, the server calls the destroy(). It stops all connections, and removes the servlet from browser's address, and closes it.

```
public void destroy() {
    - - - -
}
```

$\boxed{\text{init()}} \rightarrow \boxed{\text{service()}} \rightarrow \boxed{\text{destroy()}}$

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~~7) → JSP elements:~~

3) → GIT is a version control system.
→ Git is Distributed version Control system.
→ It stores and manages code, files, and helps in many ways:
→ Features of GIT are:

Branching: GIT allows diff developers to branch from same window and merge it later.

Collaboration: GIT allows different developers to work together on same code.

Retrieving: If any error is made, GIT allows to go back to previous version.

tracking: GIT tracks work of every single change and saves it in detail.

Conflict Revolution: If two or more developers are working on same project and error happens, GIT can point out error easily.

→ GIT repository is used to store our project software, track all the updates, upload it, share it to other developers.
→ steps to create a GIT repository:

i) login access:

git --global config user.name "sathvik"
git --global config user.email "rudsat@gmail.com"

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→ Now, we create a directory for required project.

mkdir project

cd project.

→ Now, initialize it to git as repo

git init

→ To create a file, we use:

echo "Hello" > ReadMe.md

now, ReadMe.md file is created.

→ To save this file, first add it to staging area.

git add ReadMe.md

or, add all files using git add .

→ Now, perform commit (with comment if any)

git commit -m "First commit"

→ to check status, use status

git status

→ to get history of all our activity, use log

git log

→ to get oneline history, use

git onelog

→ Now, to store it to github, create an account in github and upload it with same directory name. After creating same directory, upload it to github with commands given in github.

→ Creating a Git repo is very useful as it supports efficient version control and collaboration.

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It ensures that the code /software is of managed version of same versions. and it maintains record of every update.

→ Git repo also supports collaboration, as it allows multiple users to work on the same project at a time. It maintains every single change, maintains a log of it, and allows the developer to merge them later if needed.

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